WORKSHEET · SCRATCH (GR 6–8) BEGINNER: CODE FOUNDATIONS

Lesson 8: Decisions and Boolean Logic

Module 3 | 45 minutes | Selection and Boolean logic | Grades 6–8 · ages 11–14

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. selection ___ A. The value where a decision changes.
b. Boolean ___ B. Ways to combine or flip conditions.
c. AND / OR / NOT ___ C. A value that is true or false.
d. boundary ___ D. Choosing which code runs.

2. Plan your code

Before you build it: Grade checker: ask “Your score?” and set score to answer. If score > 89 say “Grade A”; else if score > 69 say “Grade B”; else say “Keep practising”. Write or draw the blocks you will use.

3. Plan your code

Before you build it: Ticket machine: ask “Your age?”. If age < 12 OR age > 64, say “Discount ticket”; otherwise say “Full price”. Write or draw the blocks you will use.

4. Is the condition TRUE or FALSE when age = 15?

Write each item in the right group: True / False.

a. age > 12 and age < 20 → _____
b. age < 12 or age > 64 → _____
c. not (age = 15) → _____
d. age > 10 or age > 100 → _____
e. age > 15 → _____

5. Explain your thinking

A theme park ride needs riders taller than 120 cm AND older than 7. Write the condition, and list three test cases (including a boundary) with the expected result.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can use if-else and chains of decisions.
☐ I can combine conditions with AND, OR and NOT.
☐ I can test boundaries.